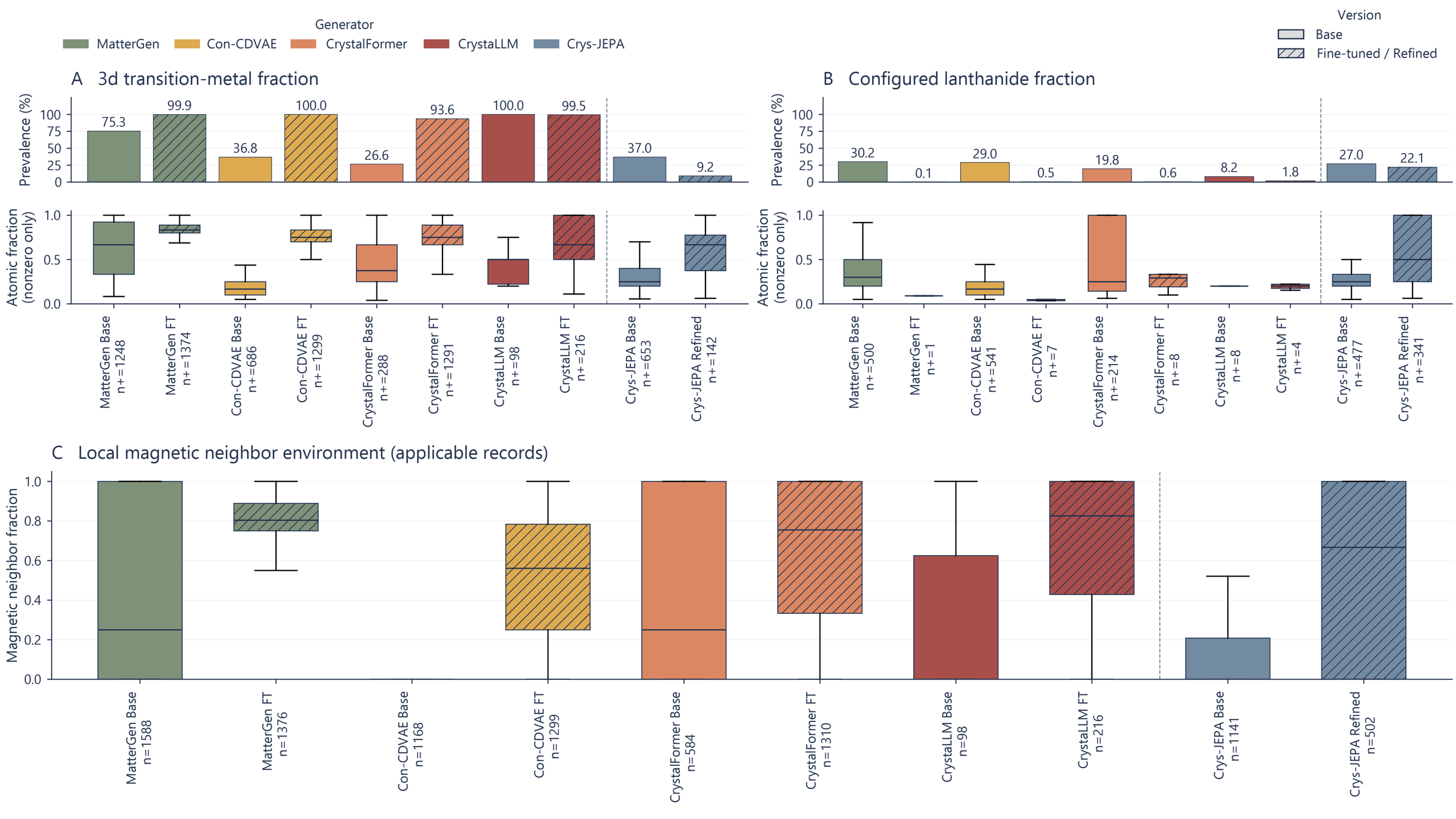

Generator
MatterGen
Con-CDVAE
CrystalFormer
CrystaLLM
Crys-JEPA
Version
Base
Fine-tuned / Refined
A 3d transition-metal fraction
Prevalence (%)
75.3
99.9
36.8
100.0
26.6
93.6
100.0
99.5
37.0
9.2
Atomic fraction (nonzero only)
MatterGen Base n+=1248
MatterGen FT n+=1374
Con-CDVAE Base n+=686
Con-CDVAE FT n+=1299
CrystalFormer Base n+=288
CrystalFormer FT n+=1291
CrystaLLM Base n+=98
CrystaLLM FT n+=216
Crys-JEPA Base n+=653
Crys-JEPA Refined n+=142
B Configured lanthanide fraction
Prevalence (%)
30.2
0.1
29.0
0.5
19.8
0.6
8.2
1.8
27.0
22.1
Atomic fraction (nonzero only)
MatterGen Base n+=500
MatterGen FT n+=1
Con-CDVAE Base n+=541
Con-CDVAE FT n+=7
CrystalFormer Base n+=214
CrystalFormer FT n+=8
CrystaLLM Base n+=8
CrystaLLM FT n+=4
Crys-JEPA Base n+=477
Crys-JEPA Refined n+=341
C Local magnetic neighbor environment (applicable records)
Magnetic neighbor fraction
MatterGen Base n=1588
MatterGen FT n=1376
Con-CDVAE Base n=1168
Con-CDVAE FT n=1299
CrystalFormer Base n=584
CrystalFormer FT n=1310
CrystaLLM Base n=98
CrystaLLM FT n=216
Crys-JEPA Base n=1141
Crys-JEPA Refined n=502